Hailo SW Suite 2025-10 (for Hailo-8/8L)

Version

2025-10 (for Hailo-8/8L)

Suite Installation

The suite can be installed in one of three ways:

- Using all-in-one docker file (see: Docker installation)
- Using all-in-one self extracted executable (see: Self Extracted Executable)
- Manual installation of the packages in a defined order (see: Manual installation)

For interaction with Hailo Devices, a physical connection is required (although not required for installation):

- PCIe interface for the evaluation board
- M.2 connector for the M.2 board
- mPCIe connector for the mPCIe board

For the full compatibility table, see Release versions compatibility.

Docker installation

Note

For more information about Docker containers, see Working With Docker Containers.

System requirements

In case of suite installation as a Docker file, the following are required:

1. Ubuntu 20.04/22.04, 64 bit
2. 16+ GB RAM (32+ GB recommended)
3. Docker package, either docker.io 20.10.07 (from Ubuntu repo), or docker-ce 20.10.6 (from Docker website).

Add the user to the docker group with the following steps:

1. Add your user to the docker group:

```
sudo usermod -aG docker ${USER}
```

2. Log out and log in.

In case you want to use Nvidia GPUs for hardware emulation and quantization, you also need to install:

1. Nvidia's Pascal/Turing/Ampere GPU architecture (such as Titan X Pascal, GTX 1080 Ti, RTX 2080 Ti, or RTX A4000)
2. GPU driver version 525
3. nvidia-docker2. It can be installed by running the following commands:

```
distribution=$(. /etc/os-release;echo $ID$VERSION_ID) \
&& curl -s -L https://nvidia.github.io/nvidia-docker/gpgkey | sudo apt-key
add - \
&& curl -s -L \
https://nvidia.github.io/nvidia-docker/$distribution/nvidia-docker.list \
| sudo tee /etc/apt/sources.list.d/nvidia-docker.list

sudo apt-get update
sudo apt-get install -y nvidia-docker2
sudo systemctl restart docker
```

Running the docker file

1. Download the suite from Developer Zone
2. Download PCIe driver from Developer Zone
3. Install PCIe driver by running the following command:

```
sudo dpkg -i <pcie_driver>.deb
```

4. Reboot the computer
5. Extract the suite archive, then run the script - which opens a new container, and attaches to it ("gets inside it"):

```
unzip hailo_ai_sw_suite_<version>.zip
./hailo_ai_sw_suite_docker_run.sh
```

It is possible to enable/disable HailoRT features inside the Suite Docker. To see available flags, run:

```
./hailo_ai_sw_suite_docker_run.sh --help
```

For example, to enable the HailoRT **multi-process service**, run:

```
./hailo_ai_sw_suite_docker_run.sh --hailort-enable-service
```

6. Run *pip list | grep hailo* to see the Hailo packages that are installed in the activated virtual environment
7. Run *hailo -h* to see a list of the available CLI commands of Hailo Dataflow Compiler and HailoRT

You can exit the docker by using the *exit* command, and then attach to it using the commands shown on Working With Docker Containers.

Another option is to use the script's additional options:

- **Resume** – go back to the existing container:

```
./hailo_ai_sw_suite_docker_run.sh --resume
```

- **Override** – delete the existing container and create new one:

```
./hailo_ai_sw_suite_docker_run.sh --override
```

Note

We have created a shared folder between the host system and the docker system. The path inside the docker is `/local/shared_with_docker/`, and the path outside is `./shared_with_docker` folder that is created on the same directory where hailo_ai_sw_suite_docker_run.sh is run.

Note

To validate that the PCIe driver was installed successfully, run *lspci | grep Co-processor*. For more information refer to *HailoRT User Guide / Installation / Validating the PCIe driver was successfully installed on Linux*.

ocker Suite Upgrade

1. Copy all private files from the currently used container in case those will be needed in the new container.
2. Make sure to close previously activated container.
3. Follow the Docker installation section in order to set up the new Hailo AI SW Suite docker container.
4. Copy the previously copied files from the previously used container to the newly set up container.

Self Extracted Executable

Hailo AI SW Suite self extracted executable will install all SW suite components directly into the system.

> **Note**
>
> TAPPAS component requires a dedicated flag.

System requirements

The following requirements are needed for the installation:

1. Ubuntu 20.04/22.04, 64 bit
2. 16+ GB RAM (32+ GB recommended)
3. Python 3.10/3.11/3.12, including `pip` and `virtualenv`
4. `python3.X-dev` and `python3.X-distutils` (according to the Python version), `python3-tk`, `graphviz`, and `libgraphviz-dev` packages. Use the command `sudo apt-get install PACKAGE` for installation
5. build-essential package (needed for compiling the PCIe driver)
6. For TAPPAS: `ffmpeg`, `x11-utils`, `python-gi-dev`, `libgirepository1.0-dev`, `libzmq3-dev` apt packages,

   `git`, `Opencv4`, `Gstreamer`, `pygobject`, `opencv_flann`, `calib3d` and `features2d`

11. (Optional) bison, flex, libelf-dev and dkms packages (needed to register the PCIe driver using DKMS)
12. (Optional) cmake (needed for compiling the HailoRT examples and for the TAPPAS installation

    when using the self extracted executable)

13. (Optional) Node.js (minimum version: v20.9.0)- required for the DFC Studio

To install TAPPAS apt packages, run the following:

```
sudo apt-get install -y ffmpeg x11-utils libgstreamer-plugins-base1.0-dev python-
gi-dev libgirepository1.0-dev libzmq3-dev
```

To install OpenCV:

```
# Download Opencv and unzip
wget https://github.com/opencv/opencv/archive/4.5.2.zip
unzip 4.5.2.zip

# cd and make build dir
cd opencv-4.5.2
mkdir build
cd build

# Make and install
cmake -DOPENCV_GENERATE_PKGCONFIG=ON \
    -DBUILD_LIST=core,imgproc,imgcodecs,calib3d,features2d,flann \
    -DCMAKE_BUILD_TYPE=RELEASE \
    -DWITH_PROTOBUF=OFF -DWITH_QUIRC=OFF \
    -DWITH_WEBP=OFF -DWITH_OPENJPEG=OFF \
    -DWITH_GSTREAMER=OFF -DWITH_GTK=OFF \
    -DOPENCV_DNN_OPENCL=OFF -DBUILD_opencv_python2=OFF \
    -DINSTALL_C_EXAMPLES=ON \
    -DINSTALL_PYTHON_EXAMPLES=ON \
    -DCMAKE_INSTALL_PREFIX=/usr/local  ..
make -j4
sudo make install

# Update the linker
sudo ldconfig
```

To install Gstreamer:

```
sudo apt-get install -y libcairo2-dev libgirepository1.0-dev libgstreamer1.0-dev \
libgstreamer-plugins-base1.0-dev libgstreamer-plugins-bad1.0-dev gstreamer1.0-
plugins-base \
gstreamer1.0-plugins-good gstreamer1.0-plugins-bad gstreamer1.0-plugins-ugly
gstreamer1.0-libav \
gstreamer1.0-doc gstreamer1.0-tools gstreamer1.0-x gstreamer1.0-alsa gstreamer1.0-
gl \
gstreamer1.0-gtk3 gstreamer1.0-qt5 gstreamer1.0-vaapi gstreamer1.0-pulseaudio \
python-gi-dev
```

For further information, please refer to Gstreamer installation guide

To install pygobject:

```
sudo apt install python3-gi python3-gi-cairo gir1.2-gtk-3.0
```

```
sudo apt-install python3-gi python3-gi-cairo gir1.2-gtk-3.0
```

For further details, please refer to pygobject installation guide

The following additional requirements are needed for GPU based hardware emulation:

1. Nvidia's Pascal/Turing/Ampere GPU architecture (such as Titan X Pascal, GTX 1080 Ti, RTX 2080 Ti, or RTX A4000)
2. GPU driver version 525
3. CUDA 11.8
4. CUDNN 8.9
5. (Recommended for TensorFlow) AVX instructions support on CPU

Installation

The executable is an archive that contains all required Hailo SW and an embedded installation script. The file will be named according to the following convention:

```
hailo_ai_sw_suite_<version>.run
```

For example:

```
hailo_ai_sw_suite_2025-10.run
```

After the executable is launched, the following will be present on the system:

- A new virtual environment named **"hailo_venv"** will be created and will contain the following:
    1. HailoRT Python package and its dependencies
    2. DFC Python package and its dependencies
    3. ModelZoo Python package and its dependencies
    4. All Python packages specified in the included requirements.txt file
    5. A target directory will be created containing all suite contents
- Hailo docs will be present in the directory from which the executable was launched
- Hailo PCIe driver installed

'sage:

First, make sure the file is executable by running `chmod +x`. If not, turn it into an executable

First, make sure the file is executable by running `ls -l`. If not, turn it into an executable with:

```
sudo chmod 770 hailo_ai_sw_suite_<version>.run
```

The following command will extract the archive contents to `hailo_ai_sw_suite` directory and will install Hailo AI SW suite:

```
./hailo_ai_sw_suite_<version>.run
```

Example:

```
./hailo_ai_sw_suite_2025-10.run
```

For those who only want to extract and keep archive contents:

```
./hailo_ai_sw_suite_<version>.run --noexec
```

To run an internal validation hook after the installation is completed:

```
./hailo_ai_sw_suite_<version>.run -- validate-installation
```

To install TAPPAS, specify the `install-tappas` flag at the end of the command:

```
./hailo_ai_sw_suite_<version>.run -- install-tappas
```

To install HailoRT and enable the **multi-process service**, add:

```
./hailo_ai_sw_suite_<version>.run -- activate-hailort-service
```

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